

FLM: Learned Choices in a Physical Feedback Loop

Pose feedback, delayed decisions and a scripted reference

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10 September 2026 — working research note

Abstract

We connect existing 256-neuron sensory-choice networks to a physical fly’s current position and heading. A declared assay crosses four fixed neural checkpoints with live or frozen pose and three waypoint scenarios, adding a scripted reference for 27 conditions. Every recorded sensory input, neural frame and delayed command passes independent replay; an additional physical repeat matches exactly. Of nine trained-model/live-pose comparisons, 9 produce the same recorded body trajectory as the scripted reference. This establishes a working feedback interface, while exposing the contribution of the engineered sensor and gait wrapper. It does not establish an advantage from anatomy, language knowledge or learned gait.

1 Question and relation to FLM

The earlier physical-choice assay computed each neural action before running the body. Here current body pose determines subsequent sensory cues during MuJoCo simulation. The question is whether the learned cue mapping works in this closed high-level loop, and how removing online pose changes tracking.

The network contains 256 central MaleCNS neurons, 6,678 directed edges and 32 readout pools [2]. It uses four sensory inputs and two action outputs; it is separate from the 1,024-neuron WikiText and BabyLM language models. Male brain data and the female NeuroMechFly body represent different specimens. This is a rate-network engineering assay, with no reconstructed visual pathway, biophysical membrane units, language weights or online learning.

2 Fixed models and physical interface

All neural checkpoints come from training seed 17 in the published cue-learning study: the shared initialization, and update 900 for full backpropagation through time (BPTT), fixed-core readout training, and supervised forward eligibility. The initial tensors agree across all three methods. The fixed-core condition trains 258 readout/normalization parameters; the others can train 8,729 total parameters. Local eligibility is an approximation motivated by prior recurrent credit-assignment work [1]; it is not unique to fly wiring.

The physical model runs on flat ground through the established FlyGym hybrid turning controller [4, 3]. Its designed gait, adhesion and contact corrections command 42 actuated joint degrees of freedom at a 0.1 ms physics step. All trials share the calibrated spawn and physical seed 17. A virtual waypoint is fixed at (40, 20) mm, fixed at (40, −20) mm, or changes from the former to the latter at $t = 1$ s. Every trial lasts two simulated seconds. Waypoints are coordinates, not visible objects in the camera.

3 Causal clock and controls

At the start of cycle k , the sampled thorax position (x_k, y_k) and yaw ψ_k define the wrapped target-bearing error

$$e_k = \text{wrap}_{[-\pi, \pi)} [\text{atan2}(y_k^* - y_k, x_k^* - x_k) - \psi_k]. \quad (1)$$

Nonnegative error encodes cue 0; negative error encodes cue 1. Each cycle has eleven 5 ms neural frames: two cue frames, eight blank frames and one query. Fast and slow state reset each cycle, matching episodic training. The prior motor command is held through the 50 ms cue-to-query delay; the first cycle starts with $[1, 1]$. Action 0 selects descending drive $[0.4, 1.2]$ and action 1 selects $[1.2, 0.4]$. When $|e_k| \leq 0.10$ rad, the shared engineered deadband instead selects straight drive $[1, 1]$. This gate uses the sampled error, not a newer pose at query time.

Live mode samples current pose each cycle. Frozen mode repeatedly uses the initial pose while preserving the current target schedule and low-level gait feedback. The scripted reference chooses the sign-of-error action directly, with the same delay, deadband and motor interface. Its three live-pose cases have no neural state or invented probabilities.

4 All physical conditions

The primary metric integrates absolute true bearing error over the complete two seconds using trapezoidal integration of the 1 ms observations. Table 1 reports degrees for readability; stored records use radians. No trial is discarded and no population confidence interval is estimated from one seed.

Controller	Pose	Positive target	Negative target	Target switch
Untrained	Live	101.48	92.14	117.09
Untrained	Frozen	101.48	93.91	117.09
BPTT	Live	7.68	7.26	16.27
BPTT	Frozen	96.41	93.91	100.50
Fixed core	Live	7.68	7.26	16.27
Fixed core	Frozen	96.41	93.91	100.50
Eligibility	Live	7.68	7.26	16.27
Eligibility	Frozen	96.41	93.91	100.50
Scripted reference	Live	7.68	7.26	16.27

Table 1: Mean absolute target-bearing error in degrees. All 27 conditions, displayed as nine controller/pose rows across three scenarios. Lower is better.

Controller	Positive target	Negative target	Target switch
Untrained	+0.00	-1.77	+0.00
BPTT	-88.72	-86.65	-84.23
Fixed core	-88.72	-86.65	-84.23
Eligibility	-88.72	-86.65	-84.23

Table 2: Live minus frozen mean error, in degrees. Negative differences favor online pose feedback conditional on the common sensor/deadband/gait wrapper.

Across all 27 conditions, there are 7 distinct recorded generalized-position trajectories. Equality is tested on all saved position arrays, separately from neural states and logits. These conditions are not 27 independent motor skills or population replications.

5 Recorded trajectories and neural state

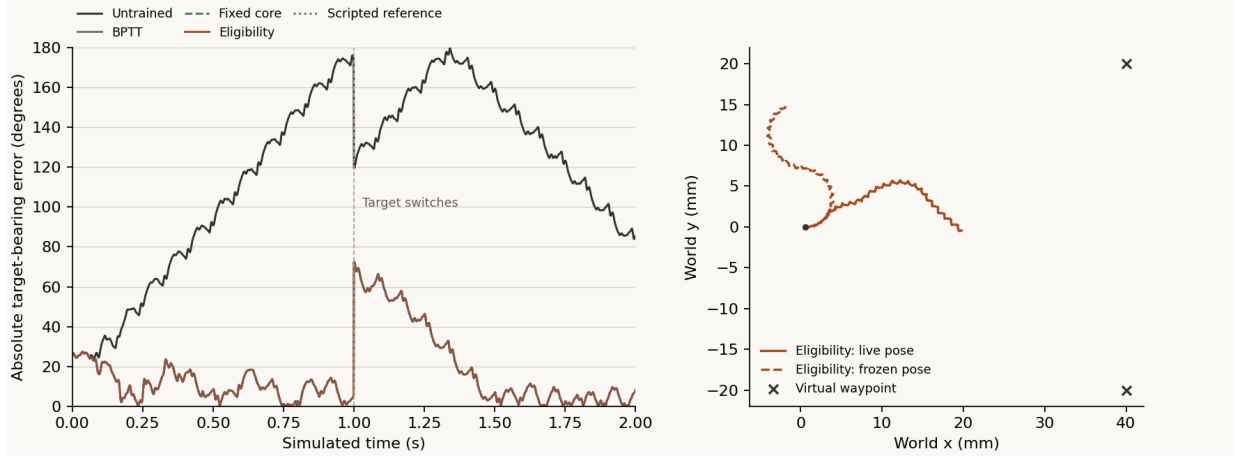


Figure 1: Target-switch scenario. Left: all five live-pose error curves; identical trajectories overlap. Right: live and frozen pose for the eligibility network. The target changes at one simulated second.

For the predeclared eligibility/live/switch video trial, mean angular error is 16.27 degrees and final target distance is 28.11 mm. Distance progress is 10.85 mm in the first phase and 10.91 mm in the second. Each phase uses its own fixed target for both endpoint distances, avoiding an artificial progress jump when the target switches.

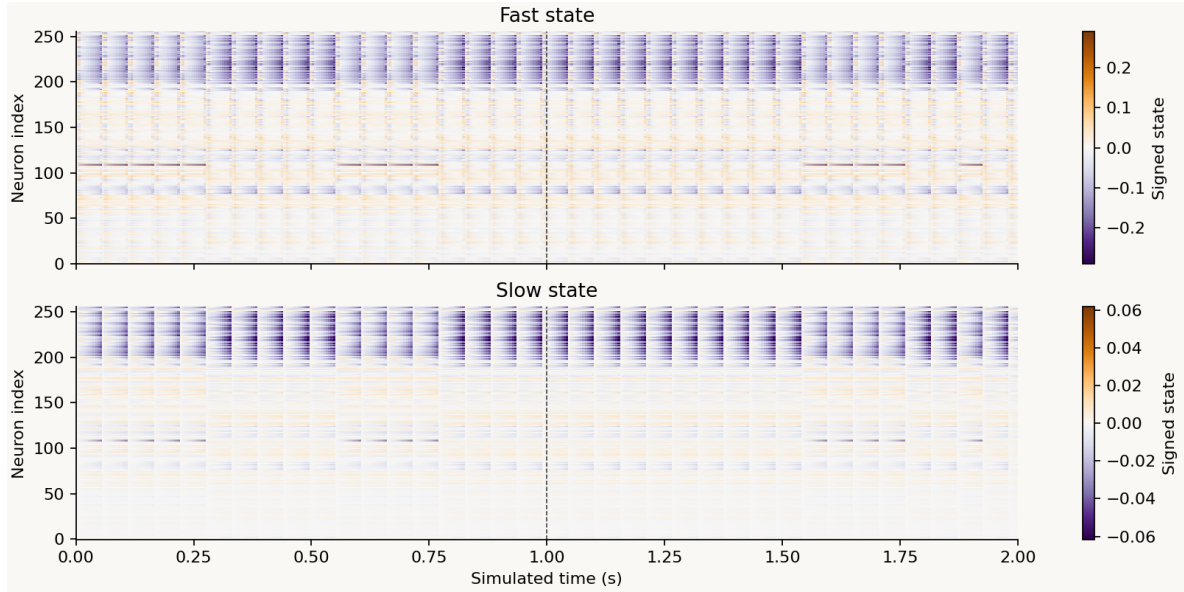


Figure 2: Actual 256-neuron state during that trial. Each column holds the inference state for the following 5 ms interval. Separate signed color scales show fast and slow variables. Eleven-frame resets match the trained episode interface; these panels do not establish continuous long-term memory.

The video contains 200 frames at 25 fps, displaying two simulated seconds in eight seconds. Its tracking camera follows the body; virtual target coordinates are shown in the trajectory plot. The repeated trial reproduces every physical and neural array and all delayed decisions exactly in the recorded environment.

6 Independent verification and reproduction

The NumPy bridge exports the effective normalized recurrence, time constants, input projection, pooling, LayerNorm and readout from the original PyTorch checkpoints. Four fixed 128-frame fixtures include random sensory inputs, resets and the original zero-distractor cue episodes. Fast/slow state and logits pass combined absolute tolerance 2×10^{-6} and relative tolerance 2×10^{-5} in both environments; argmax choices agree. This is numerical parity within tolerance, not bitwise equivalence between all arithmetic kernels.

The independent auditor recomputes pose-derived sensory inputs and replays all 400 control frames and 36 delayed decisions per condition. It reconstructs the applied motor timeline and recomputes all reported physical measurements. The main cohort contains 54,027 body observations, 10,800 control frames and 972 decisions; the separate repeat adds 2,001 observations, 400 frames and 36 decisions. Scripted control frames contain sensory and command records only.

Five fault-injection checks alter cue, query timing, neural state, sampled pose or phase progress. Each edited archive has a refreshed valid hash; the auditor still rejects its causal or numerical inconsistency. Complete-cohort reporting refuses to publish a comparison until all conditions and the repeat pass.

The public records archive includes every NPZ/CSV/JSON record, four controllers, parity fixtures, source files, graph identities, protocols, licenses and a payload hash manifest. From a fresh extraction, run `python scripts/audit_feedback_release.py`. Auditing saved records needs NumPy, without PyTorch or MuJoCo. Fresh physics additionally uses the pinned FlyGym 2.1.0/MuJoCo 3.9.0 environment and the documented empty-directory workflow; its identity must remain distinct from the published observations.

7 What the assay establishes

The sensory mapping can be exercised in a causal body-feedback loop. The trained-controller/scripted trajectory comparisons reveal when the engineered interface makes different internal models behaviorally indistinguishable. Matching the fixed-core or scripted reference provides no evidence that trained recurrent dynamics were necessary for this task.

The conditions were declared before physical execution, but methods were chosen from previously observed cue-learning results. The experiment is exploratory. Ground-truth position, sign encoding, an externally prescribed target, episodic resets, a straight-ahead gate and a designed gait controller all simplify the problem. The single seed cannot support population claims. No model learns from physical reward, controls individual joints through learned weights, or transfers language knowledge to behavior here.

Anatomical utility remains a separate hypothesis requiring matched retrained wiring controls. The ongoing WikiText study has priority over larger-corpus training. A later language-to-control study needs an explicit shared interface and matched adaptation of untrained and language-trained cores; this physical assay cannot substitute for that comparison.

References

- [1] Guillaume Bellec et al. A solution to the learning dilemma for recurrent networks of spiking neurons. *Nature Communications*, 2020. <https://doi.org/10.1038/s41467-020-17236-y>.
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